

OSTROVNOE, Russian Federation

WMO: 251380

Lat: 68.112N

Lon: 164.173E

Elev: 310

StdP: 14.53

Time Zone: 12.00 (E12)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-51.9	-49.0	-57.2	0.2	-50.5	-54.5	0.2	-49.0	21.8	26.2	19.0	9.7	1.2	80	0.786

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	79.4	60.3	75.6	59.0	71.9	57.7	63.2	74.9	60.9	71.3	58.7	69.2	5.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	73.7	67.5	55.8	67.1	66.2	53.3	61.2	64.0	28.8	75.6	27.2	71.5	25.7	69.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.4	14.2	12.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-55.0	86.3	2.7	4.1	-56.9	89.2	-58.5	91.6	-60.0	93.9	-61.9	96.8
			-55.0	67.0	2.7	3.8	-56.9	69.8	-58.5	72.0	-60.0	74.1	-61.9	76.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	-26.0	-25.0	-7.2	14.9	37.2	54.6	57.1	50.7	38.9	16.9	-5.2	-21.1
	DBStd	33.18	18.14	15.17	16.49	13.32	10.52	8.05	8.52	7.42	7.85	11.99	16.20	17.33
	HDD50	13088	2357	2100	1774	1054	412	48	30	81	347	1025	1657	2203
	HDD65	18031	2822	2520	2239	1504	863	316	272	446	784	1490	2107	2668
	CDD50	565	0	0	0	0	14	187	249	103	12	0	0	0
	CDD65	35	0	0	0	0	0	6	25	4	0	0	0	0
	CDH74	490	0	0	0	0	0	118	326	45	1	0	0	0
	CDH80	99	0	0	0	0	0	6	91	2	0	0	0	0
Wind	WSAvg	4.9	4.1	3.7	5.0	5.2	6.1	5.8	5.5	5.1	5.2	4.8	4.4	4.0
Precipitation	PrecAvg	9.4	0.4	0.4	0.4	0.2	0.5	1.2	1.5	1.5	1.1	0.8	0.7	0.6
	PrecMax	11.7	0.9	1.2	1.1	0.5	1.1	2.5	2.8	3.0	2.9	1.8	1.5	1.3
	PrecMin	6.7	0.1	0.0	0.0	0.0	0.0	0.4	0.2	0.4	0.2	0.2	0.1	0.2
	PrecStd	1.3	0.2	0.3	0.2	0.1	0.3	0.5	0.7	0.7	0.6	0.4	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	31.8	37.7	45.3	70.0	81.6	87.3	78.7	69.3	45.6	33.9	32.8
		MCWB	31.6	28.4	33.2	36.8	53.3	57.7	66.8	61.6	54.2	39.8	30.3	29.6
	2%	DB	28.6	20.7	29.9	41.7	63.7	77.9	82.3	74.5	62.1	40.2	30.9	23.7
		MCWB	25.0	18.6	25.6	34.3	50.1	57.0	63.8	59.3	51.2	37.3	28.1	21.6
	5%	DB	11.6	3.3	22.5	38.7	58.3	74.4	77.2	70.0	56.2	35.7	24.7	13.3
		MCWB	10.3	2.3	19.5	32.1	46.9	56.6	61.3	57.5	48.0	33.4	22.9	12.1
	10%	DB	0.6	-5.9	16.0	35.3	53.9	70.8	72.8	65.6	51.3	31.8	19.1	3.9
		MCWB	-0.3	-6.7	13.3	29.8	43.8	55.6	59.5	55.0	45.9	30.3	17.7	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.8	28.3	33.3	37.5	55.1	62.1	69.6	63.9	55.8	40.3	30.8	30.0
		MCDB	36.8	31.8	37.5	44.8	66.8	75.2	85.1	75.4	65.9	44.3	33.2	32.4
	2%	WB	24.9	17.9	25.8	34.6	50.1	59.3	64.9	61.1	51.9	37.2	27.8	21.6
		MCDB	28.7	19.8	29.7	41.2	62.4	71.8	77.9	71.5	60.0	39.9	30.8	23.6
	5%	WB	10.4	1.6	19.7	32.4	47.3	57.7	62.5	58.6	49.1	33.6	22.8	11.9
		MCDB	11.2	2.3	22.9	38.4	58.3	71.2	74.3	68.4	55.0	35.7	24.7	13.2
	10%	WB	-0.3	-6.6	13.3	29.8	44.1	56.1	60.4	56.0	45.9	30.4	17.7	3.2
		MCDB	0.5	-6.1	15.8	35.0	53.4	69.2	70.4	63.7	50.7	31.8	19.1	3.9
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.6	19.8	22.5	17.4	19.3	18.4	19.0	14.7	11.0	12.2	12.1
		MCDBR	23.4	20.1	16.7	18.9	24.2	26.5	26.8	27.7	25.6	11.5	13.6	20.4
		MCWBR	21.2	18.0	14.4	13.6	13.9	10.9	12.9	14.3	15.2	8.5	11.9	19.2
	5% WB	MCDBR	23.5	18.5	16.8	17.8	22.9	23.0	24.5	24.8	22.2	10.9	13.9	20.3
		MCWBR	21.4	17.5	14.4	12.9	13.6	11.0	13.5	13.4	13.9	8.0	12.3	19.1
Clear-Sky Solar Irradiance	taub	0.165	0.222	0.294	0.344	0.380	0.365	0.416	0.398	0.320	0.257	0.183	N/A	
	taud	1.923	2.086	2.061	1.975	2.127	2.387	2.238	2.245	2.435	2.161	1.921	N/A	
	Ebn at Noon	42	187	228	247	254	265	242	232	229	165	27	0	
	Edh at Noon	3	15	28	41	40	32	36	32	20	13	2	0	
All-Sky Solar Radiation	RadAvg	10	173	718	1481	1834	1806	1562	1152	607	220	31	0	
	RadStd	1	18	35	44	167	122	147	121	52	22	3	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)		+3.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page